

Strength Training and Cardiovascular Exercise

Core principles, types, intensity, and recovery for building a balanced exercise routine

Introduction to Strength Training

Strength training, also called resistance training, involves working muscles against a force to make them stronger over time. It targets major muscle groups, including the legs, back, chest, shoulders, arms, and core.

Principles of Strength Training

Progressive Overload

Muscles adapt and grow stronger when they are gradually challenged with more weight, more repetitions, or more difficult movements over time.

Frequency and Rest

The CDC recommends strength training for all major muscle groups on 2 or more non-consecutive days per week, allowing muscles time to recover between sessions that target the same muscle group.

Proper Form

Using correct technique reduces the risk of injury and helps target the intended muscles effectively. Learning movements with lighter loads before increasing weight is a common and safe approach.

Types of Strength Training

- Bodyweight exercises: push-ups, squats, lunges, planks
- Free weights: dumbbells, barbells, kettlebells
- Resistance bands: portable and adjustable resistance
- Machines: guided movement patterns often found in gyms

Introduction to Cardiovascular (Aerobic) Exercise

Cardiovascular exercise, or aerobic exercise, raises the heart rate and breathing rate over a sustained period. It strengthens the heart and lungs and improves the body's ability to use oxygen efficiently.

Types of Cardio Exercise

- Walking or brisk walking
- Running or jogging
- Cycling

- Swimming
- Dancing
- Group fitness classes

Measuring Cardio Intensity

Heart rate can be used as a general guide to exercise intensity. A commonly used estimate for maximum heart rate is 220 minus your age. Target heart rate zones are often expressed as a percentage of this estimated maximum:

- Moderate intensity: roughly 64–76% of estimated maximum heart rate
- Vigorous intensity: roughly 77–93% of estimated maximum heart rate

The talk test described in fitness fundamentals guides is a simple alternative that does not require tracking heart rate.

Balancing Strength and Cardio in a Weekly Routine

A sample way to structure a balanced week (adjust based on personal ability and goals):

- 2–3 days of strength training covering major muscle groups
- 3–5 days including moderate or vigorous aerobic activity
- At least 1 full rest or light-activity day

Recovery Between Workouts

Muscles repair and strengthen during rest, not during the workout itself. Training the same muscle group on consecutive days without rest can slow progress and increase injury risk. Mild muscle soreness after new or intense exercise is common and typically resolves within a few days.

Safety Tips for Strength and Cardio Training

- Warm up before and cool down after every session
- Increase intensity or load gradually rather than suddenly
- Allow rest days, especially after strength training the same muscles
- Stop and rest if you feel sharp pain, dizziness, or unusual shortness of breath
- Consult a healthcare provider before starting if you have an existing health condition

Sources

- U.S. Centers for Disease Control and Prevention (CDC) — Muscle-Strengthening Activities and Physical Activity Guidelines

- World Health Organization (WHO) — Guidelines on Physical Activity and Sedentary Behaviour
- National Institutes of Health (NIH) — MedlinePlus, Exercise and Physical Fitness
- UK National Health Service (NHS) — Strength and Flexibility Exercises, Aerobic Exercise

This document provides general educational information only. It is not medical advice, diagnosis, or treatment. Consult a qualified healthcare provider for individual health concerns.